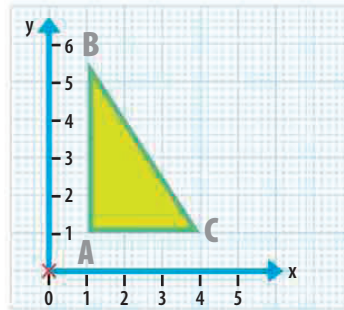


LOOKING CLOSER

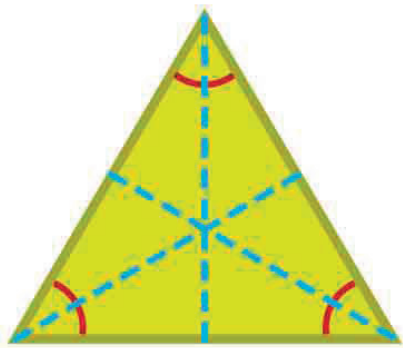
Graphs and geometry

Graphs link geometry with other areas of mathematics. Plotting lines and shapes in graphs with coordinates makes it possible to convert them into algebraic expressions, which can then be manipulated mathematically. The reverse is also true: algebraic expressions can be shown on a graph, enabling them to be manipulated using the rules of geometry. Graphical representations of objects enables positions to be given to them, which makes it possible to apply vectors and calculate the results of movements, such as rotations and translations.



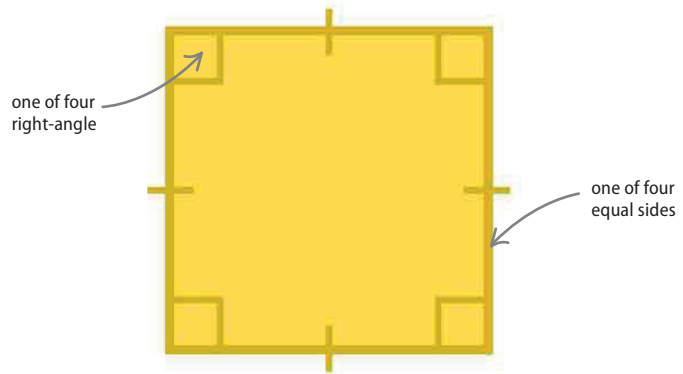
◁ Graph

The graph here shows a right triangle, ABC, plotted on a graph. The vertices (corners) have the coordinates $A = (1, 1)$, $B = (1, 5.5)$, and $C = (4, 1)$.



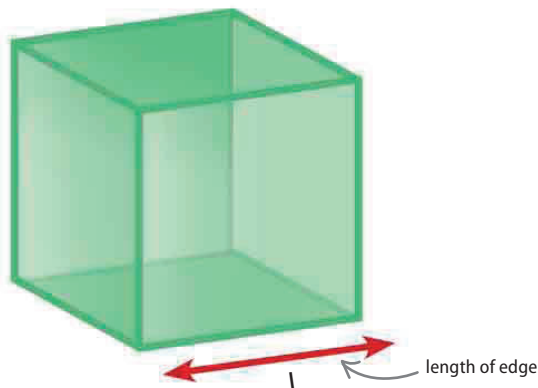
△ Triangle

A triangle is a three-sided, two-dimensional polygon. All triangles have three internal angles that add up to 180° .



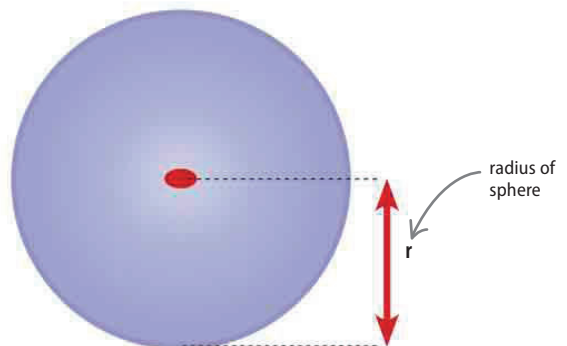
△ Square

A square is a four-sided polygon, or quadrilateral, in which all four sides are the same length and all four internal angles are right angles (90°).



△ Cube

A cube is a three-dimensional polygon in which all its edges are the same length. Like other rectangular solids, a cube has 6 faces, 12 edges, and 8 vertices (corners).



△ Sphere

A sphere is a perfectly round three-dimensional shape in which every point on its surface is the same distance from the center; this distance is the radius.